External Peripheral Interfaces

Course Teacher:

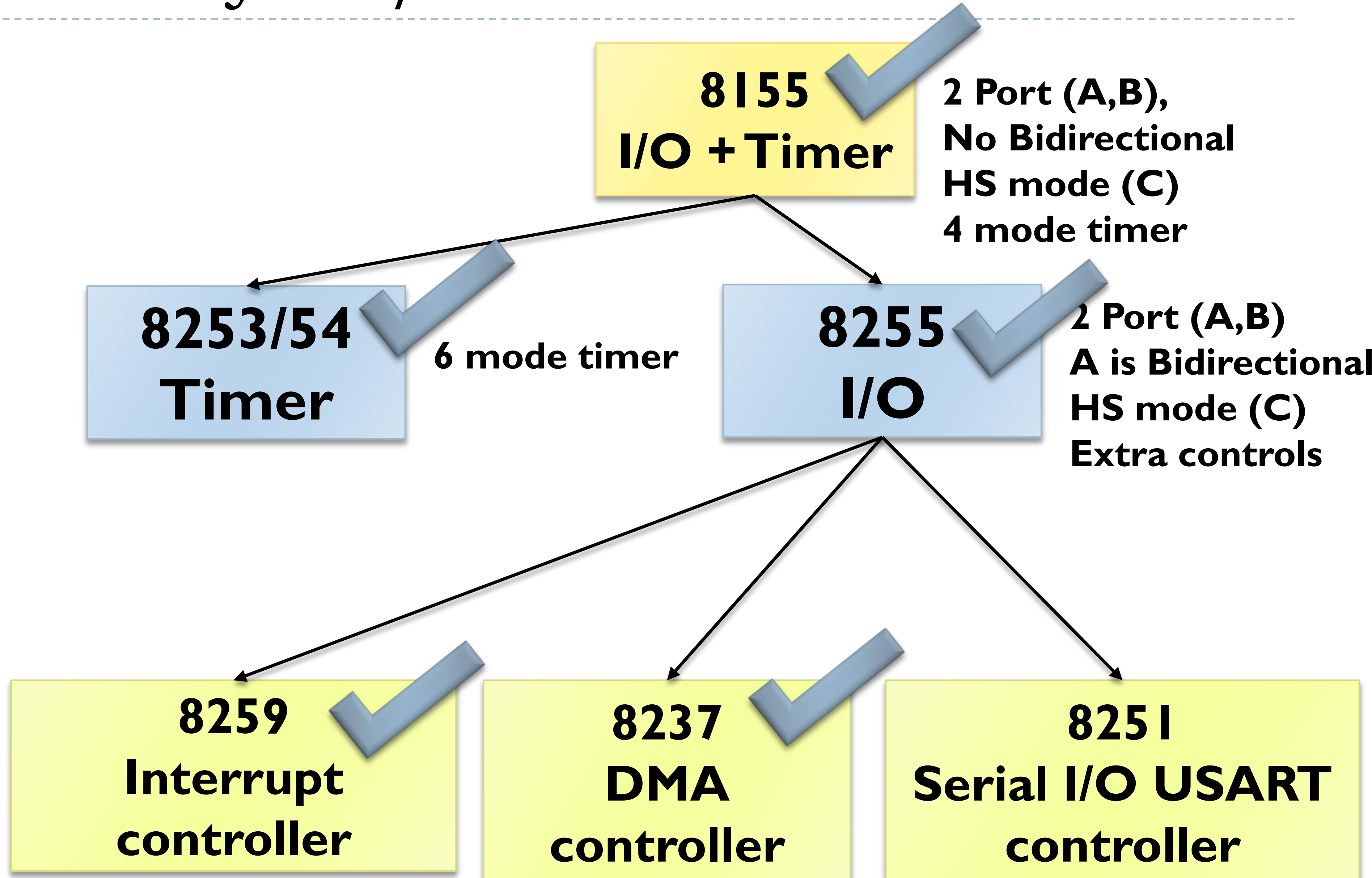
Md. Obaidur Rahman, Ph.D.
Professor
Department of Computer Science and Engineering (CSE),
Dhaka University of Engineering & Technology (DUET), Gazipur.

Course ID: CSE - 4619
Course Title: Peripherals, Interfacing and Embedded Systems
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Lecture References:

- Book:
  - *Microprocessor Architecture, Programming and Applications with 8085 (Chapter-15)*, **Author:** Ramesh Gaonkor
  - *Microprocessors and Interfacing: Programming and Hardware*, **Author:** Douglas V. Hall
- Lecture Materials:
  - *I/O System Design*, Dr. Esam Al_Qaralleh, CE Department, Princess Sumaya University for Technology.

Hierarchy of I/O Control Devices



External Interface Concepts

- Recalled: OSI model – level 1
  - Physical level, which requires peripheral devices to connect two different computers or devices together
  - Termed as "Interface"
- Two types of External Interfaces Standards
  - **EIA-232F** (RS-232 Serial Interface/COM Port)
  - **USB** (Universal Serial Bus)
- Other peripheral interfacing standards that provide power, flexibility and ease-of-installation include:
  - **FireWire** (low cost device for digital)
  - **SCSI** (mainly for permanent storage, CD/DVD) or iSCSC
  - **HDMI** (for Video and Gaming Display)
  - **InfiniBand**, **Fibre Channel** (high speed connection)

UART/USART IC

- Writing a program compatible with all different serial communication protocols is difficult and it is an inefficient use of microprocessor.
- **UART:** Universal Asynchronous Receiver/Transmitter chip.
  - RS-232 Serial port uses UART
- **USART:** Universal Synchronous/Asynchronous Receiver/Transmitter chip
- The microprocessor sends/receives the data to the UART/USART in parallel, while with I/O, the UART/USART transmits/receive data serially.

8251 UART/USART IC

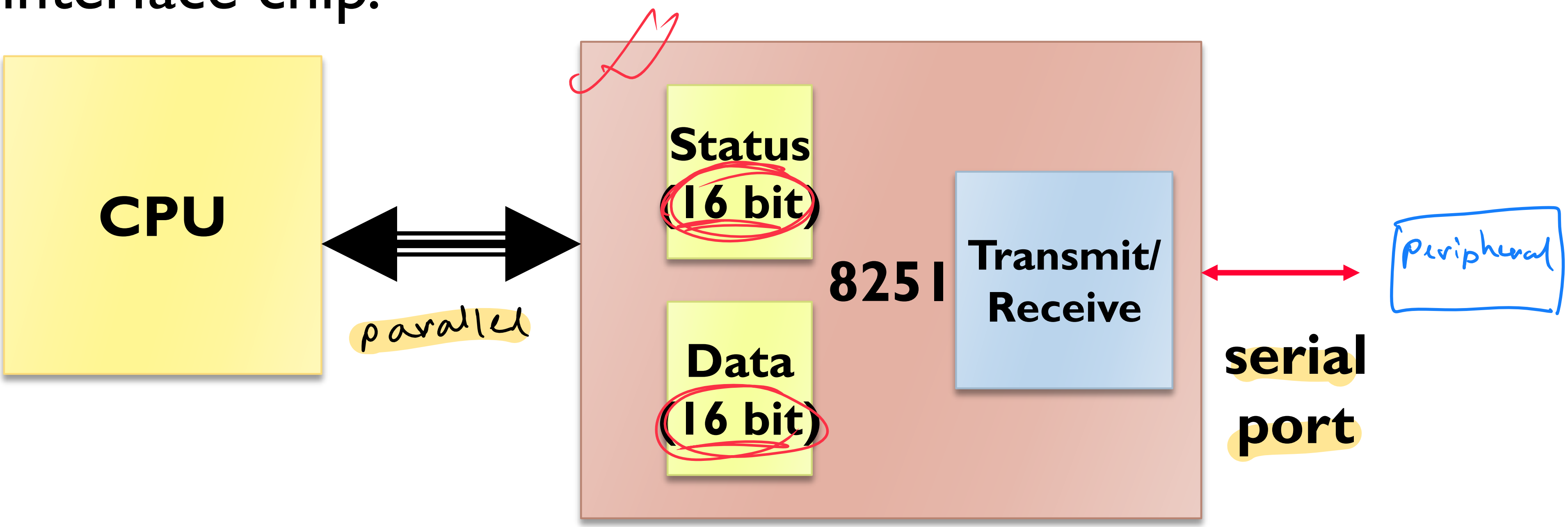
- 8251 universal synchronous asynchronous receiver transmitter (USART) acts as a mediator between microprocessor and peripheral to transmit serial data into parallel form and vice versa.
  - It takes data serially from peripheral (outside devices) and converts into parallel data.
  - After converting the data into parallel form, it transmits it to the CPU.
  - Similarly, it receives parallel data from microprocessor and converts it into serial form.
  - After converting data into serial form, it transmits it to outside device (peripheral).

8251 UART/USART IC

- The 8251A is a programmable serial communication interface chip designed for synchronous and asynchronous serial data communication.
- It supports the serial transmission of data.
- It is packed in a 28 pin DIP.

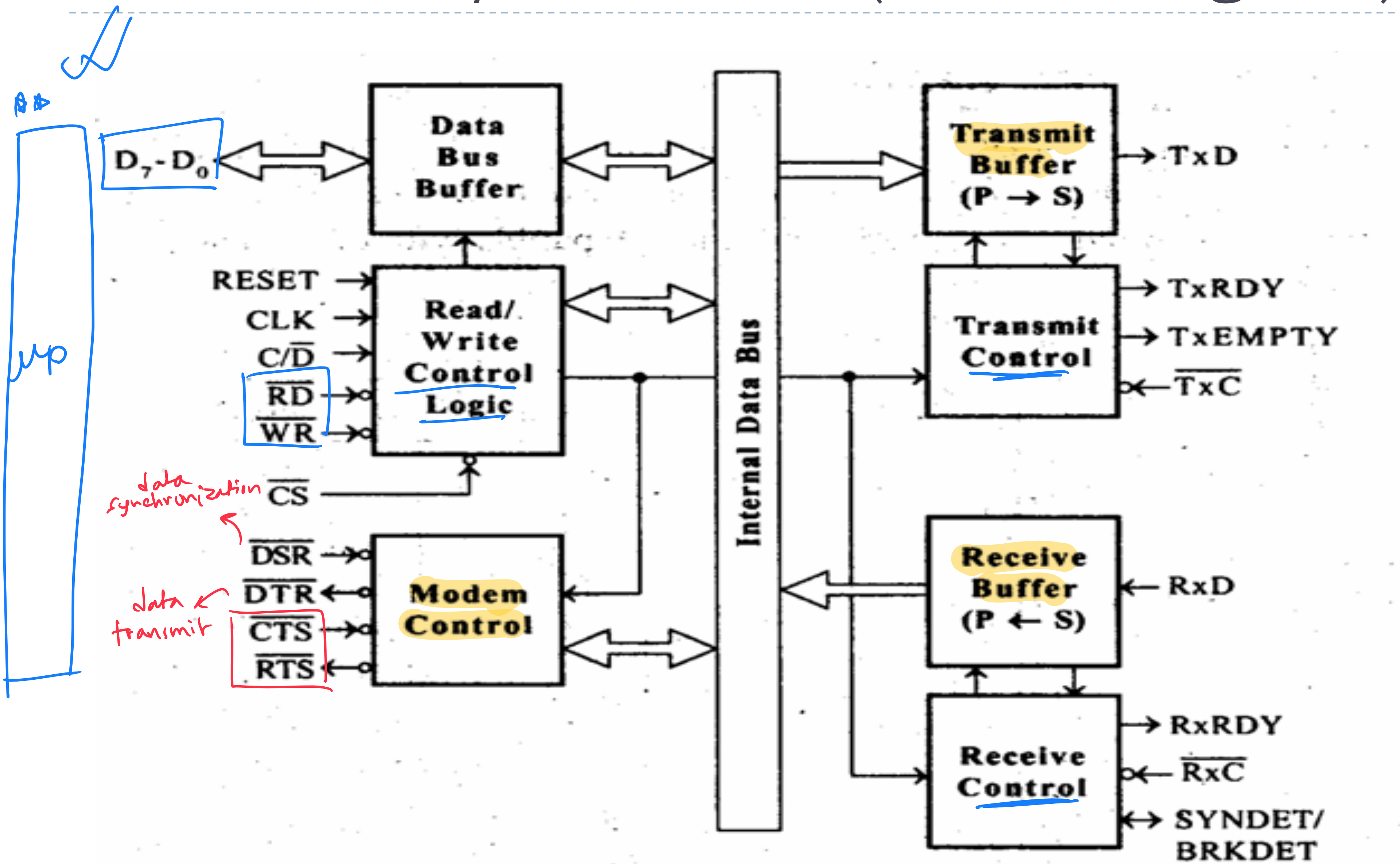
8251 UART/USART IC

- 8251 functions are integrated into standard PC interface chip.

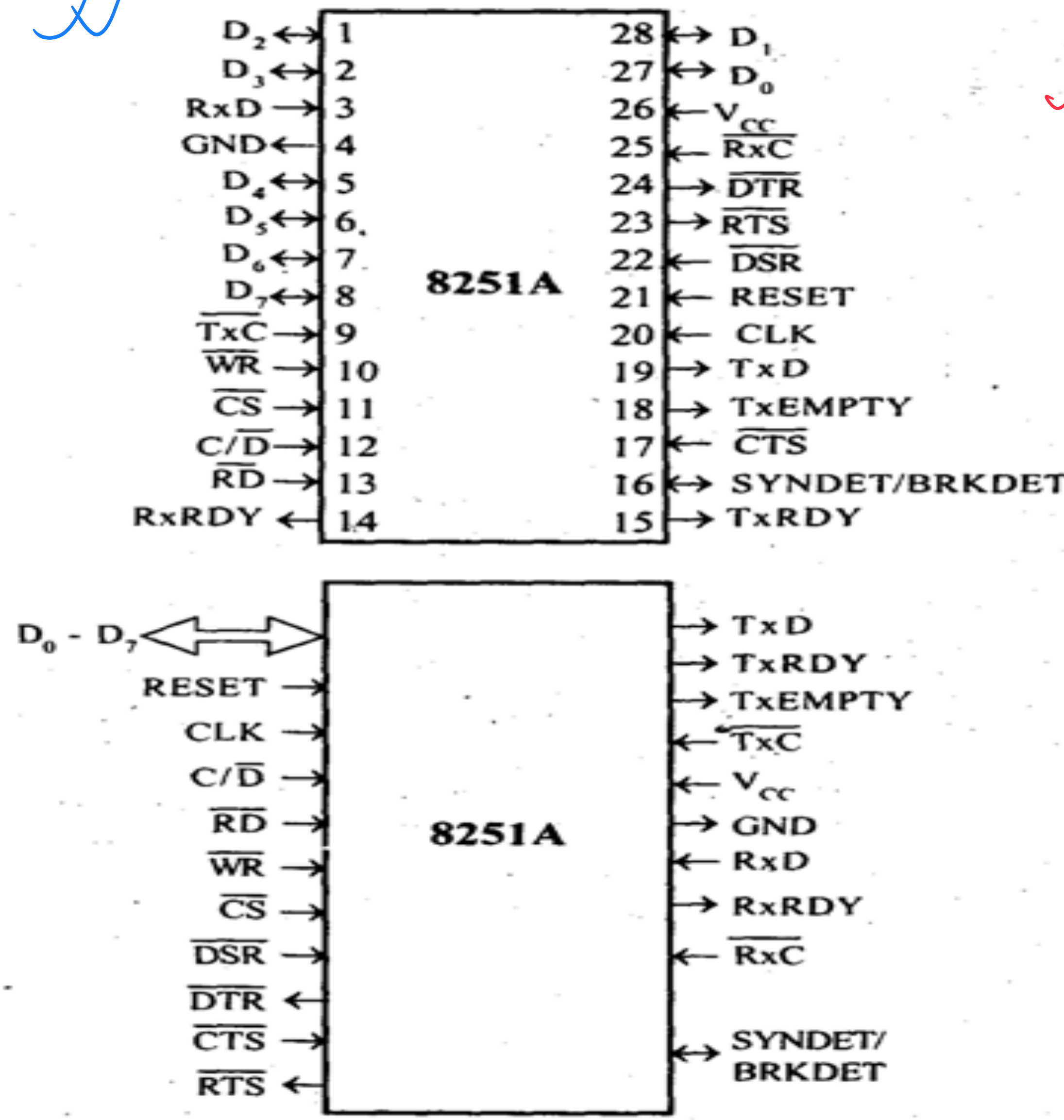


- UART/USART
  versions
  - 8251 USART
  - 8250/16450 UART is a newer version of 8251
  - 16550 is the latest version UART

8251 UART/USART IC (Block Diagram)



8251 UART/USART IC (Pin Diagram)



Pin	Description
D_0-D_7	Parallel data
$C/\overline{D}$	Control register or Data buffer select
$\overline{RD}$	Read control
$\overline{WR}$	Write control
$\overline{CS}$	Chip Select
CLK	Clock pulse (TTL)
RESET	Reset
$\overline{TxC}$	Transmitter Clock
TxD	Transmitter Data
$\overline{RxC}$	Receiver Clock
RxD	Receiver Data
RxRDY	Receiver Ready
TxRDY	Transmitter Ready
$\overline{DSR}$	Data Set Ready
$\overline{DTR}$	Data Terminal Ready
SYNDET/ BRKDET	Synchronous Detect / Break Detect
$\overline{RTS}$	Request To Send Data
$\overline{CTS}$	Clear To Send Data
TxEMPTY	Transmitter Empty
V_{cc}	Supply (+5V)
GND	Ground (0 V)

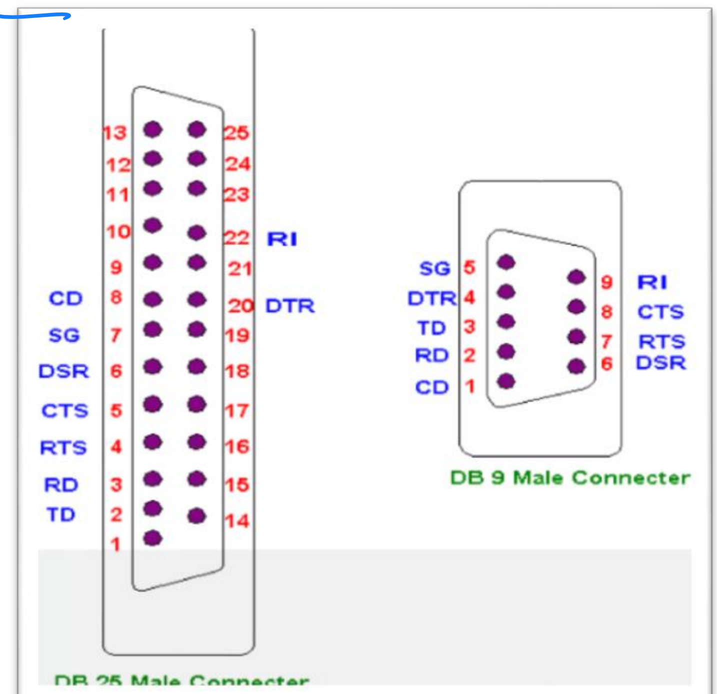
8251 UART/USART IC

Feature	UART	USART
Communication Modes	Asynchronous only	Synchronous and Asynchronous
Clock Signal	No dedicated clock signal	Provides an external clock (XCK pin) for synchronous communication
Synchronization	Relies on start and stop bits for timing	Uses a shared clock signal for precise synchronization
Data Rate	Slower and lower maximum bit rates due to timing variations	Faster and more reliable data transmission
Data Transfer	Transmits data in bytes, one bit at a time	Can transfer data in blocks during synchronous operation
Complexity & Flexibility	Simpler to implement but less flexible	More complex due to the clock but offers greater flexibility and compatibility
Physical Interface	Typically a two-wire (Tx/Rx) interface	Can utilize four wires (Tx, Rx, XCK, XDIR) for synchronous operation

Why UART?

- *A UART may be used when:*
  - High speed is not required
  - An inexpensive communication link between **two** devices is required

- *UART communication is very cheap*
  - Single wire for each direction (plus ground wire)
    - Asynchronous because no clock signal is transmitted
  - Relatively simple hardware



Advantages of USART/UART

- **Versatility:** The 8251 USART can be used for both synchronous and asynchronous communication, making it a versatile peripheral.
- **Error Detection:** The USART includes built-in error detection features, such as parity checking, which help to ensure the accuracy of transmitted data.
- **Flow Dontrol:** The USART includes flow control features, which allow for the regulation of data transmission and reception, preventing data loss and overloading.
- **Compatibility:** The 8251 USART is compatible with a wide range of microprocessors, making it a popular choice for serial communication in many different systems.
- **Ease of Use:** The USART includes simple interface pins and registers, making it relatively easy to use and program.

Disadvantages of USART

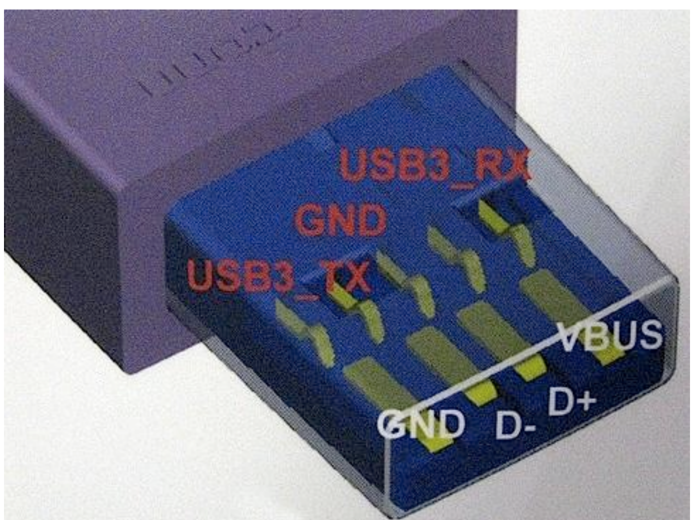
- **Limited Speed:** The 8251 USART has a relatively low maximum data transfer rate of 115.2 kbps, which may be insufficient for some applications.
- **Limited Buffer size:** The USART has a small internal buffer size, which may result in data loss if data is not read from the buffer in a timely manner.
- **Complex Programming:** Although the interface pins and registers of the USART are relatively simple, programming the USART can be complex, requiring careful attention to timing and other parameters.
- **Cost:** While the 8251 USART is relatively affordable, it does add cost to a system, particularly if multiple USARTs are required.
- **Limited Functionality:** While the 8251 USART is a useful peripheral for serial communication, it does not include more advanced features, such as DMA (direct memory access) or advanced error correction.

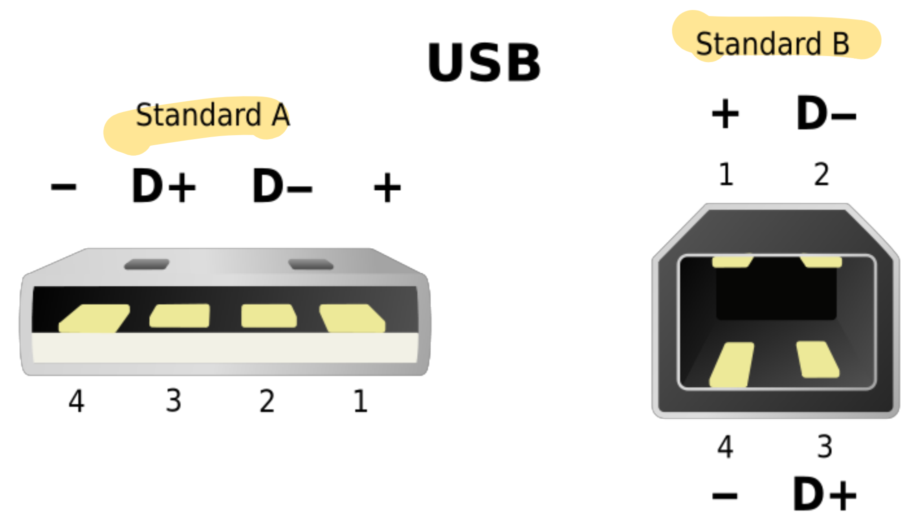
Universal Serial Bus (USB)

- A newer standard that is much more powerful than $\boxed{\text{EIA-232F}}$ multiple ports for connecting monitor/graphics
- The USB interface is a modern standard for interconnecting a wide range of peripheral devices to computers
- Supports plug and play
- Can daisy-chain multiple devices
- USB 2.0 can support 480 Mbps or 60 MBps (USB 1.0 is only 12 Mbps);
  mega bits per second mega Bytes per second
- USB 3.0 ?? (5 Gbps or 640 MBps)

Universal Serial Bus (USB)

- The USB interface defines all four components
  - The **electrical component** defines two wires VBUS and Ground to carry a 5-volt signal, while the D+ and D- wires carry the data to/from and signaling information.
  - The **mechanical component** precisely defines the size of four different connectors and uses only four wires (the metal shell counts as one more connector)





Universal Serial Bus (USB)

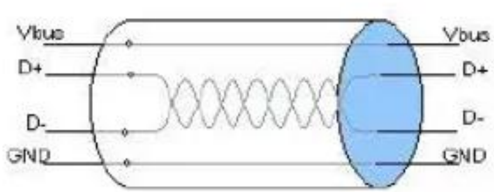
- This is now a very common interface for use with printers, scanners, digital cameras and it can also be used for keyboards and mice.
- It allows "hot swapping" which means that you can plug and unplug it while the computer is on.
- You can theoretically attach up to 128 USB devices at the same time using hubs in a "chain".
- The USB connection also provides power to the devices.

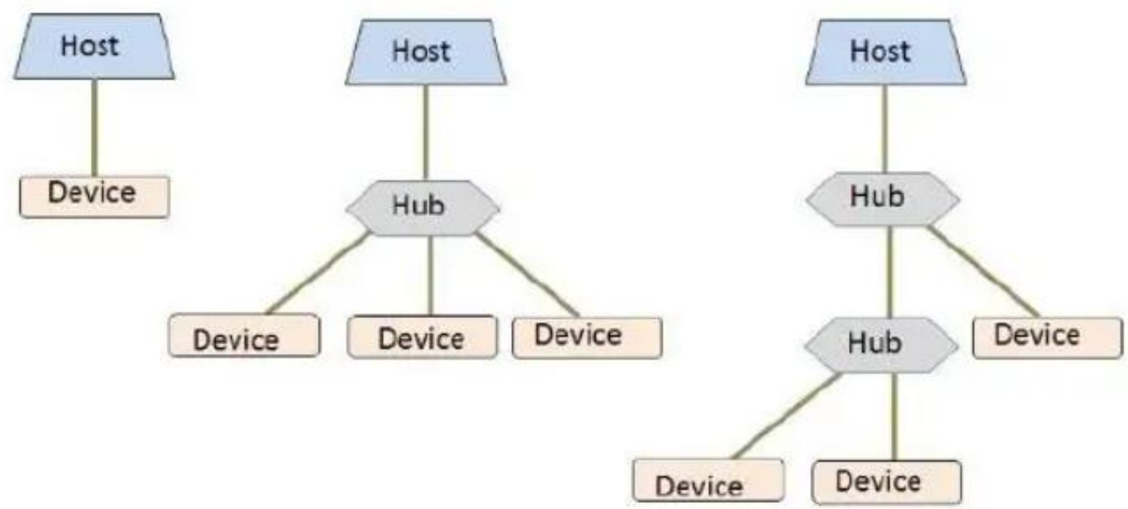
USB 1 transfer speed 12 megabits per second.
USB 2 transfer speed 480 megabits per second.
USB 3 transfer speed 5 gigabits per second.

Universal Serial Bus (USB)

- A USB system is a unidirectional communications hierarchy consisting of a single USB Host controller.
- All components of a USB system connect to each other using one of the USB specified Cables and connectors.
- USB signals are transmitted using differential signaling on a twisted-pair data cable (D+ and D-)

daisy chain





Universal Serial Bus (USB)

Version	Speed modes	Released	Data rates	Types	Power Delivery	Application
1.0	Low Speed	1996	1.5 Mbps	Half Duplex	2.5W	Keyboard, Mouse, Game Peripherals
1.1	Full speed		12 Mbps			Scanner, Printer, Digital audio
2.0	High speed	2000	480 Mbps			Broadband, Mass storage, Imaging
3.0	Super speed	2008	5 Gbps	Full Duplex	4.5~7. 5W	Real Time Streaming, Portable Storages Devices
3.1	Super speed plus	2013	10 Gbps			
3.2	SS + (with Multi link mode)	2017	20 Gbps		100W	

Universal Serial Bus (USB)

Advantages

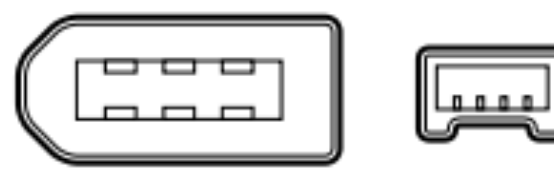
- Ease of Use
- Single Interface for multiple devices
- Auto-configuration
- Easy to expand
- Compact Size
- Speed
- Reliability
- Low cost
- Low power consumption

Limitations

- Distance
- Broadcasting.
- Each USB device can't have more than 5 "nodes" between it and the PC
- Speed compared with other protocols.

FireWire

- Low-cost digital interface (real time connection for PC)
- **A FireWire connection lets you send data to and from high-bandwidth digital devices such as digital camcorders, and it's faster than USB 1.0** (not than 3.0)
- Capable of supporting transfer speeds of up to 400 Mbps
- Hot pluggable
- Supports two types of data connections:
  - **Asynchronous** connection
  - **Isochronous** connection



FireWire

- FireWire (also called IEEE 1394 or i.LINK) was originally developed by Apple Computer for connecting peripheral devices to Macintosh computers.
- Companies such as Sony picked up on the technology and began incorporating FireWire into its camcorders and PCs, and FireWire grew from there.
- Some HDTV devices used FireWire connections for years, but FireWire appears to have all but gone away as a means of connecting HDTV devices.
- FireWire is becoming more common in the *audio* side of the home theater.

FireWire

- This is the standard interface for use with digital video camcorders, some storage devices and the first Apple Ipod.
- It allows hot swapping and up to 63 Firewire devices can be connected at the same time.
- It can supply modest power services to devices.

FireWire 400 transfer speed 400 megabits per second.

FireWire 800 transfer speed 800 megabits per second.

IEEE 1394b transfer speed 3.2 gigabits per second. → usb 3.0 supports 5GB

USB vs. FireWire

FEATURE	USB	FIREWIRE
Versions	2.0	800
Data transfer rate	480 Mbps	800 Mbps
Number of devices	127	63
Plug and play	Yes	Yes
Hot-pluggable	Yes	Yes
Bus power	Yes	Yes
Bus type	Serial	Serial
Cable type	Twisted pair (4 wires: 2 power, 1 twisted-pair set)	Twisted pair (8 wires: 2 power, 2 twisted-pair sets, 2 ground)
Networkable	Yes - host-based	Yes - peer-to-peer
Network topology	Hub	Daisy chain

HDMI (High Definition Multimedia Interface)

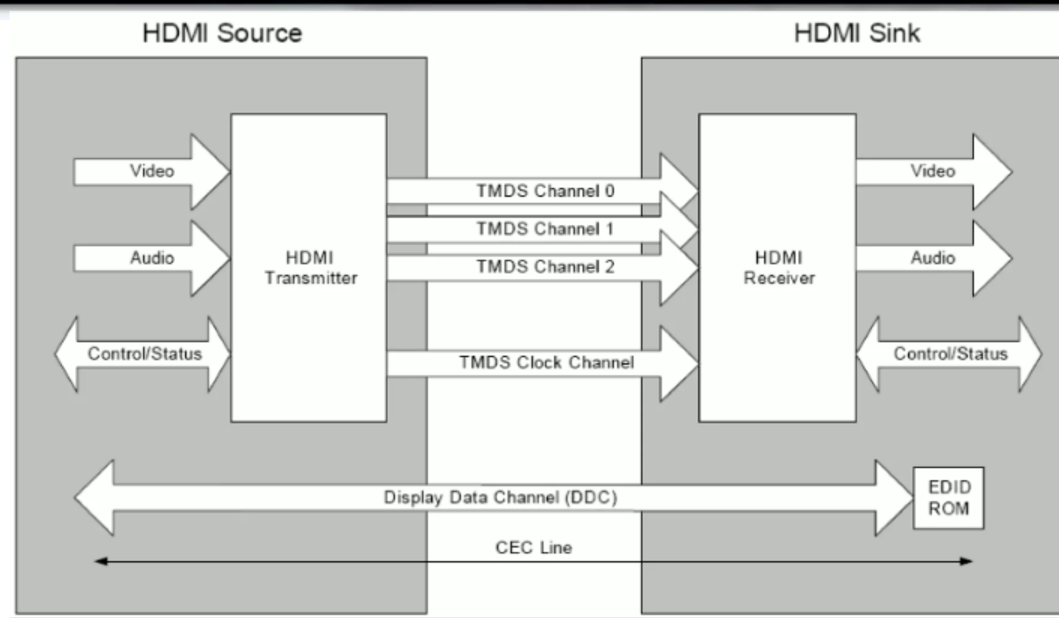
- The latest and greatest in digital video (and audio) connections is the HDMI cable.

- HDMI includes 19 wires wrapped in a single cable and able to carry a bandwidth of 18 Gbps, which is more than twice the bandwidth need to transmit multi-channel audio and video

- HDMI is included in many devices, including HDTVs, DVD players, Blu-ray disc players, cable and satellite set-top boxes, Media Center Edition PCs, and gaming consoles.

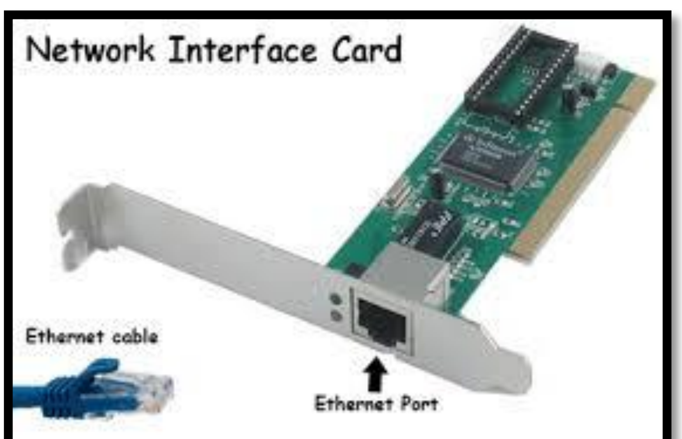
HDMI (High Definition Multimedia Interface)

HDMI VERSION	1.0	1.1	1.2	1.3	1.4	2.0
Date initially released	12/9/02	5/20/04	8/22/05	6/22/06	6/5/09	9/4/13
Maximum Bandwidth (Gbps)	4.95	4.95	4.95	10.2	10.2	18
Maximum Resolution	1600×1200p60	1600×1200p60	1600×1200p60	2048×1536p75	4096×2160p24	4096×2160p60
Maximum LPCM Audio Channels	8 Channels	8 Channels	8 Channels	8 Channels	8 Channels	32 Channels
Maximum Audio Sampling Rate	768kHz	768kHz	768kHz	768kHz	768kHz	1536kHz



SCSI legacy hardware

- **SCSI (Small Computer System Interface)**
  - A technique for interfacing a computer to high-speed devices such as hard disk drives, tape drives, CDs, and DVDs, Printers, Networks etc.
  - Designed to support devices of a more permanent nature
    - Ethernet Interface
    - Parallel Port Interface



network interfaces

InfiniBand and Fibre Channel

- **InfiniBand** – a serial connection or bus that can carry multiple channels of data at the same time
  - Can support data transfer speeds of 2.5 billion bits (2.5 gigabits) per second and address thousands of devices, using both copper wire and fiber-optic cables
  - A network of high-speed links and connecting switches

modular switches and routers use this

- **Fibre Channel** – also a serial, high-speed network that connects a computer (i.e., as server) to multiple input/output devices
  - Supports data transfer rates up to billions of bits per second, but can support the interconnection of up to 126 devices only

NIC card uses this fibre channel interface

Thank You !!